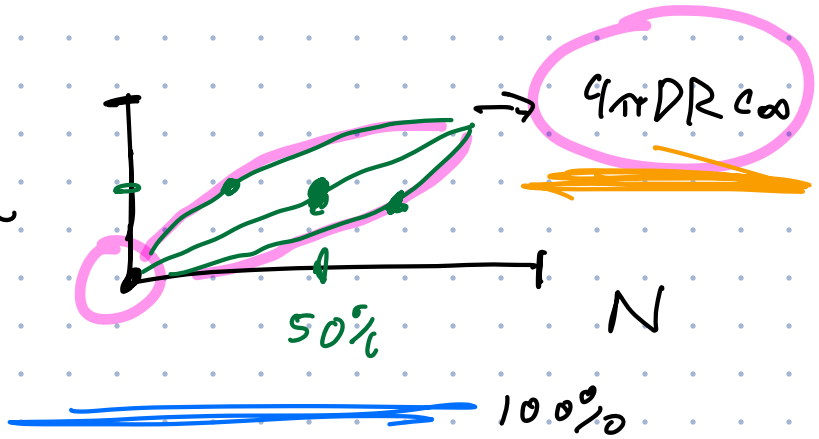
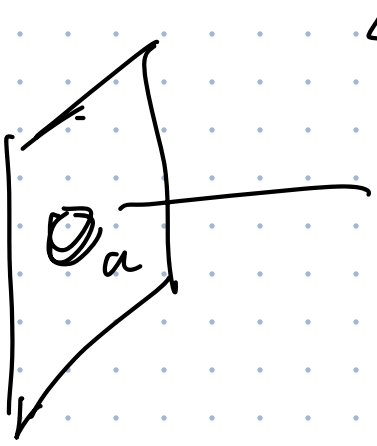
PS 4

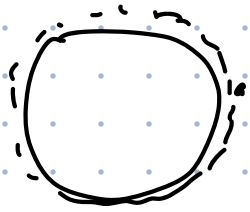


λ - ARRIVAL RATE





$\langle J \rangle \overset{?}{=} c_I, R_I$



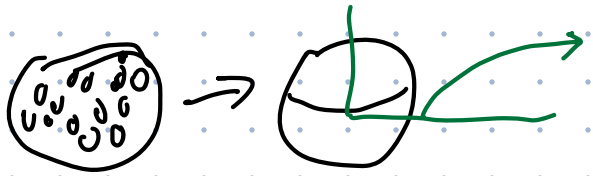
$R \approx R_I$

$J(R_I) \approx \langle J \rangle$

$c(r) =$

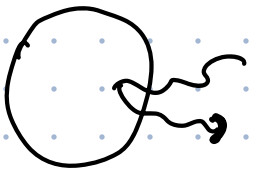
As $N \to \infty$,

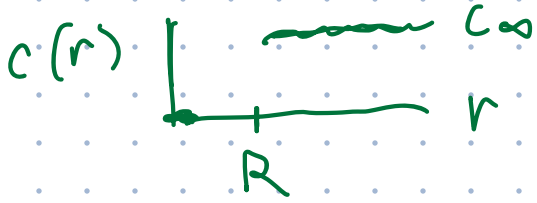
$$c(r) = c_\infty \left(1 - \frac{R}{r}\right)$$



As $N \to 0$

$$c(r) \simeq c_\infty$$





(a)

c_I

$$c(r) = c_\infty \left(\left(\frac{Na}{\pi R + Na} \right) \frac{R}{r} + 1 \right)$$

iii) $\mathcal{A} = -J A$ (A: AREA)

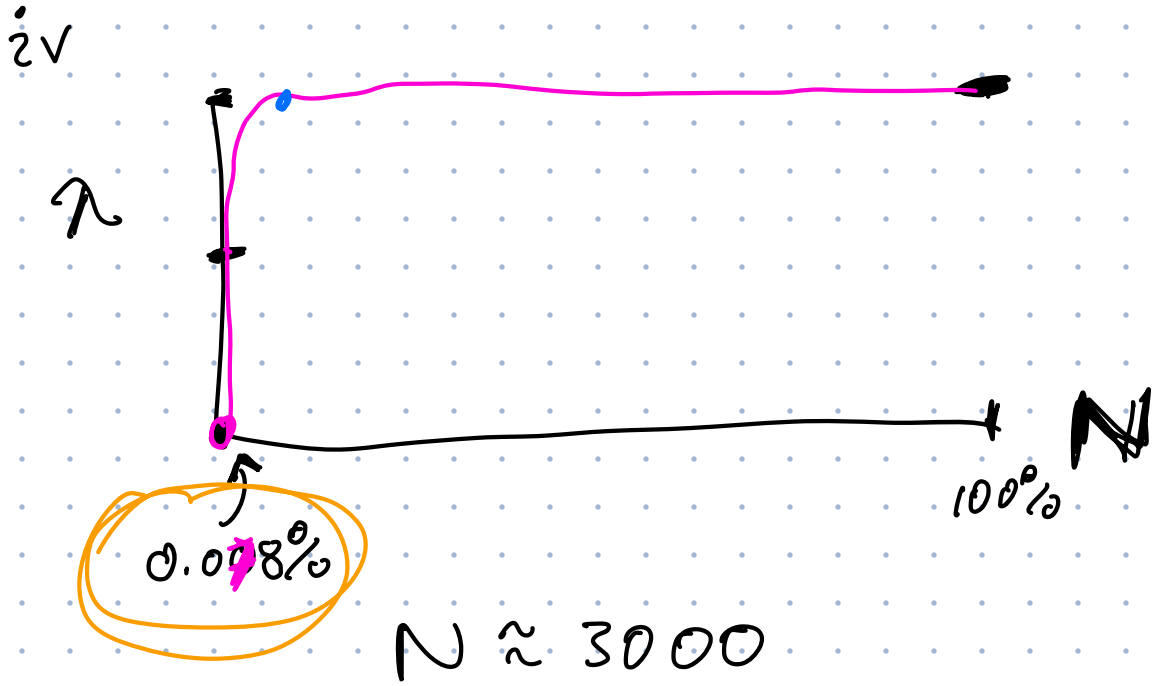
$$= \frac{D \partial c}{\partial r} 4\pi R^2$$

$$\lambda \overset{?}{=} 4DNa\left(\frac{\pi R}{\pi R + Na}\right)c_\infty$$

CHECK $N \to 0$ ✓

$N \to \infty$ ✓ $\quad \lambda = 4D\pi R\left(\frac{Na}{\pi R + Na}\right)c_\infty$

$$= 4D\pi R\, c_\infty$$



$N \approx 3000$

